



examGenome CheatSheet on Algebra

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1	What is a proportion in math? Answer: A proportion is when two ratios (or fractions) are equal. Explanation: For example, $2/4 = 3/6$ is a proportion because both equal $1/2$. It shows that two relationships are the same.
2	What is the rule for checking if two ratios form a proportion? Answer: Use cross-multiplication: if $a/b = c/d$, then $a \times d = b \times c$. Explanation: This is called the 'cross-products rule'. If the cross products are equal, the ratios are in proportion.
3	What is the difference between direct and inverse proportion? Answer: In direct proportion, when one thing increases, the other increases too. In inverse proportion, when one increases, the other decreases. Explanation: Direct: more apples = more cost. Inverse: more workers = less time to finish a job.
4	Do proportions have units? Answer: The units must match on both sides, but the proportion itself is a number with no units. Explanation: For example, $2 \text{ kg} / \$4 = 5 \text{ kg} / \10 — kilograms and dollars are used, but the ratio (0.5 kg per \$1) is unit-based. In pure proportions like $1/2 = 2/4$, there are no units.
5	What is a unit rate, and how is it used in proportions? Answer: A unit rate is a ratio where the second number is 1 (like 5 km per 1 hour). It helps compare things fairly. Explanation: Unit rates make it easy to set up proportions. Example: If a car goes 60 km in 1 hour, you can find how far it goes in 3 hours using proportion.
6	Why do we use proportions in real life? Answer: To scale recipes, read maps, compare prices, and convert measurements (like inches to centimeters). Explanation: Proportions keep relationships the same when things get bigger or smaller—like doubling a cake recipe.
7	What property allows you to solve for a missing number in a proportion? Answer: The fact that in a proportion $a/b = c/d$, you can solve for any missing value using cross-multiplication. Explanation: For example, if $3/4 = x/8$, then $3 \times 8 = 4 \times x \rightarrow 24 = 4x \rightarrow x = 6$.
8	Can proportions involve whole numbers, fractions, or decimals? Answer: Yes! Proportions can use any kind of number—as long as the two ratios are equal. Explanation: Examples: $1/2 = 0.5/1$, or $3:6 = 12:24$. The form doesn't matter—equality does.

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9	What does it mean when two things are 'in proportion'? Answer: It means their sizes or amounts change by the same scale factor—keeping the same shape or ratio. Explanation: For example, if you draw a picture twice as wide and twice as tall, it's 'in proportion' to the original.
10	What is the 'scale factor' in a proportion? Answer: The number you multiply by to get from one ratio to the other. Explanation: In $2/5 = 6/15$, the scale factor is 3 because $2 \times 3 = 6$ and $5 \times 3 = 15$. It shows how much bigger or smaller one set is compared to the other.

